

Challenge or Pass? A Dynamic Policy for MLB’s Automated Ball-Strike System

Abstract

Major League Baseball’s automated ball-strike challenge system requires teams to decide whether an uncertain correction is worth risking scarce future inventory. We combine role-specific models of observed challenge behavior, empirical pitch-margin distributions by count, and run expectancy in a dynamic decision rule. We train the policy on games before July 20 and evaluate it on 497 later ABS games. On this held-out opportunity clock, the learned policy captures 0.560 modeled run expectancy (RE) per game, compared with 0.459 under observed decisions: a gain of 0.100 RE per game (whole-game fixed-policy 95% interval 0.074–0.128). Its model-implied overturn rate is 45.3%, compared with 54.2% under observed decisions. The policy challenges more often when the immediate stake is large and future inventory is less valuable. A strong ABS policy maximizes expected run value by spending uncertain challenges where a correction matters most.

1 Introduction

In 2026, Major League Baseball (MLB) introduced the automated ball-strike (ABS) challenge system. A batter, catcher, or pitcher may contest an umpire’s call, after which the automated zone supplies the ruling. Each team starts with two challenges, retains a challenge after an overturn, and loses one after an upheld call. A club with no challenges receives one upon entering an extra inning (Major League Baseball, 2025). These rules turn a called pitch into a second decision: accept the call or risk a limited resource to correct it.

Challenge accuracy is an appealing but incomplete objective. A nearly certain overturn can have little value in a low-stakes count, while an uncertain challenge can be worthwhile when correcting the call substantially changes expected runs. Waiting also has a cost: an unused challenge cannot recover the present call, and its value declines as future opportunities disappear. The relevant comparison is therefore the expected gain from acting now against the expected value lost if the challenge fails.

Prior baseball work models called-strike probabilities and pitch framing after accounting for location, count, and participants (Deshpande and Wyner, 2017), and Bayesian work values pitch-level swing-or-take decisions under uncertainty (Yee and Deshpande, 2024). Research on line-call challenges in tennis makes challenge inventory an explicit dynamic resource (Abramitzky et al., 2012). ABS has also been studied in the Korean Baseball Organization (Lee et al., 2025), but MLB’s retain-on-success and extra-inning rules create a distinct inventory problem. Our setting joins these ideas: players observe a pitch imperfectly, the value of a correction depends on the baseball state, and failure reduces future options.

We make three contributions. First, we estimate reduced-form, role-level action-curve widths from observed challenge behavior. Second, we combine a latent noisy private signal with an empirical distribution of margins by role and count to obtain a pitch-specific success probability. Third, we learn an inventory-sensitive policy and evaluate it on a fixed sequence

of later opportunities. On the held-out factual clock, the resulting policy captures 21.8% more modeled RE than observed decisions despite a lower overturn rate because it concentrates attempts on calls for which success matters most.

2 Data and Evaluation Design

We combine MLB’s live game feed with Baseball Savant and Statcast (Major League Baseball, 2026b,a). The feed supplies pitch order, calls, challengers, and review outcomes; Statcast supplies pitch coordinates, the batter-specific zone, count, outs, base occupancy, score, and participant identities. We match sources by game, plate appearance, and physical pitch number, reconstruct the original call from the final call and review outcome, and validate challenged pitches against the official Savant publication.

The raw 2026 snapshot contains 1,989 completed games from March 25 through August 25 and 292,381 called pitches, of which 292,127 have complete tracking. The estimation sample is the 1,490-game schedule through July 19; 1,488 of those games used ABS. The held-out confirmation sample consists of 497 ABS-enabled games from July 20 through August 25 (Castrovince, 2026). The ABS-enabled development period contains 6,263 challenges; the confirmation period contains 2,275, comprising 1,233 overturned and 1,042 upheld calls. We estimate run-expectancy tables from 2023–2025 Statcast data.

Games define every cross-fitting fold and bootstrap cluster. All prior selection, width estimation, nuisance fitting, policy learning, and robustness selection use only the development sample; no confirmation game contributes to a fitted coefficient, prior, or policy choice.

3 Method

The analysis moves from revealed behavior to a probability of success and then to a value-sensitive action:



3.1 Pitch geometry and immediate value

We reconstruct the published two-dimensional ABS zone as a 17-inch-wide rectangle with batter-specific vertical bounds and evaluate the tracked pitch center at the middle of the plate (Major League Baseball, 2025; Castrovince, 2026). After subtracting a 1.45-inch baseball radius, Euclidean distance handles the four zone corners. This construction agrees with all 8,537 tracked official Savant rulings in the publication set. Let d_i be the resulting signed ball-edge distance to the boundary: $d_i > 0$ is outside and $d_i < 0$ is inside. We orient the margin toward the adversely affected team,

$$M_i = \begin{cases} d_i, & \text{if the umpire called a strike (offensive opportunity),} \\ -d_i, & \text{if the umpire called a ball (defensive opportunity).} \end{cases} \quad (1)$$

Thus $M_i > 0$ always denotes a call that public ABS geometry says is correctable. Official ABS outcomes exist only for challenged pitches; validated public geometry supplies a common adjudication rule for all opportunities.

We estimate RE288, the expected runs through the end of the half-inning for each of 8 base states, 3 out states, and 12 valid ball-strike counts. Rare states shrink toward count-independent RE24 (8 base states by 3 out states) using 2023–2024 for tuning and 2025 for validation, after which the table is refit on all three seasons. Write $u_r(x) = x$ for offense and $u_r(x) = -x$ for defense so that value is always oriented toward the challenging side. The immediate benefit of correcting pitch i is

$$G_i = u_{r_i}\{RE_i(\text{corrected call})\} - u_{r_i}\{RE_i(\text{call stands})\}. \quad (2)$$

The fixed opportunity clock preserves the realized pitch sequence and reconstructs each team’s challenge inventory event by event. The estimand is modeled RE captured on opportunities that occurred; downstream game states remain fixed at their observed values.

3.2 Revealed perception and the empirical prior

An eligible adverse call is a tracked called strike for the offense or called ball for the defense for which the relevant team has at least one challenge. Let A_i indicate a challenge and let X_i collect the observed covariates. Within development-game folds, we fit separate offense and defense probit generalized additive models. Across the full margin range, the role- r challenge propensity is

$$\begin{aligned} \Phi^{-1}\{P(A_i = 1 \mid X_i, r)\} = & \beta_{0r} + f_{M,r}(M_i) + f_{v,r}(v_i) + f_{Mv,r}(M_i, v_i) \\ & + f_{G,r}(G_i) + f_{h,r}(h_i) + f_{\Delta,r}(\Delta_i) + \mathbf{z}_i^\top \gamma_r + \eta_{j(i),r} + \tau_{t(i),r}. \end{aligned} \quad (3)$$

Here Φ is the standard-normal CDF; the f terms are smooth functions, with $f_{Mv,r}$ the margin-speed interaction. Also, v_i is release speed, h_i is inning, Δ_i is the role-oriented score margin, and $\mathbf{z}_i$ contains count, pitch family, matchup, outs, inventory, and missing-speed indicators. The final two terms are role-appropriate participant and team random effects. A companion local probit fit within three inches of the boundary uses the linear margin term $b_r M_i$ and omits the release-speed terms while retaining the other state, factor, participant, and team terms. Its effective discrimination width is $w_r = 1/b_r$. The fitted widths are 2.940 inches for offense and 1.817 for defense and set the scale of the latent-normal signal model.

The primary policy carries forward these widths, not the GAM’s fitted challenge probabilities. Policy evaluation retains every tracked called-pitch opportunity on the factual clock and forces a wait whenever inventory is zero.

A signal has different implications at different counts because the pitches presented to a challenger are different. For role r and raw count c , we place all structurally tracked adverse-call development margins on 0.01-inch bins, irrespective of challenge inventory. The cross-fitted prior probability for bin b is

$$p_{rc}(b) = \frac{n_{rc}(b) + \alpha p_r(b)}{n_{rc} + \alpha}. \quad (4)$$

Here b indexes a 0.01-inch bin, $n_{rc}(b)$ is its role-count frequency, $n_{rc} = \sum_b n_{rc}(b)$, and $p_r(b)$ is the role-wide empirical mass. The shrinkage weight $\alpha = 200$ is selected separately by role using a game-clustered held-out continuous ranked probability score and the one-standard-error rule. Challenge outcomes do not enter this prior.

Let the modeled private signal be $S_i = M_i + \varepsilon_{s,r}$, where $\varepsilon_{s,r} \sim N(0, \sigma_{s,r}^2)$. Combining the signal with the empirical prior gives

$$q_{rc}(s) = P(M > 0 \mid S = s, r, c) = \frac{\sum_{b>0} p_{rc}(b) \phi_{\sigma_{s,r}}(s - b)}{\sum_b p_{rc}(b) \phi_{\sigma_{s,r}}(s - b)}, \quad (5)$$

where ϕ_σ is a normal density with standard deviation σ . At $\sigma_{s,r} = 0$, equation (5) is interpreted by its point-mass limit, $q(s) = \mathbf{1}\{s > 0\}$.

3.3 Partial identification and the inventory rule

Observed behavior identifies the total width w_r ; its decomposition into sensory noise and threshold compliance remains unidentified. Write the action index as $\tilde{S}_i = S_i + \varepsilon_{a,r}$, with independent $\varepsilon_{a,r} \sim N(0, \sigma_{a,r}^2)$. We therefore index the unidentified split by

$$\sigma_{s,r} = \kappa_r w_r, \quad \sigma_{a,r} = w_r \sqrt{1 - \kappa_r^2}, \quad \kappa_r \in \{0, .25, .50, .75, 1\}. \quad (6)$$

Sensory noise governs the posterior; action noise governs imperfect compliance with the threshold, and independence gives $w_r^2 = \sigma_{s,r}^2 + \sigma_{a,r}^2$. For each of the 25 offense-defense pairs, we optimize a candidate's expected captured RE per team-game on the 1,490 development clocks, cross-evaluate every candidate under all 25 evaluation decompositions, and select the candidate with the largest worst-case expected RE. The selected training pair is $(\kappa_o, \kappa_d) = (1.00, 1.00)$, and its binding worst-case evaluation is $(1.00, 1.00)$. This binding scenario determines the maximin choice; it is not a structural estimate of the sensory-action noise split.

Let $C_i(k)$ denote the continuation value of holding k challenges at pitch i , and let $L_{ik} = C_i(k) - C_i(k - 1)$ be the marginal loss from a failed challenge. A success captures G_i and retains inventory; a failure captures nothing and reduces inventory by one. The policy challenges when

$$q_i G_i > (1 - q_i) L_{ik}, \quad q_i > q_{ik}^* = \frac{L_{ik}}{G_i + L_{ik}}. \quad (7)$$

We directly learn smooth, stage-specific loss surfaces subject to $L_{i1} \geq L_{i2} \geq 0$ on the fixed clocks. Extra innings replenish an empty inventory to one before the inning's opportunities. The direct maximin rule is the primary policy; an independently solved Bellman policy is a structural check.

Here stage indexes half-innings and current outs through the end of regulation; extra innings reuse six half-inning-by-outs states. The direct loss surfaces use role-specific regulation-stage splines, current-outs indicators, and an extra-inning indicator. The rule always waits when $k = 0$ or $G_i \leq 0$.

3.4 Held-out scoring and uncertainty

For held-out scoring, we integrate the fitted signal rule over possible private signals conditional on the realized margin,

$$P(A_i = 1 \mid M_i) = \int P(A_i = 1 \mid S_i = s) dP(s \mid M_i). \quad (8)$$

Equation (8) converts realized geometry into an ex post expected action rate; the live rule uses S_i , not M_i . We propagate the resulting action, success, failure, and inventory probabilities down each confirmation-game clock. Sampling uncertainty comes from 5,000 whole-game resamples of the 497-game confirmation sample with the fitted policy held fixed.

4 Results

Public geometry is complete for 72,633 of 72,653 confirmation-period called pitches and identifies 4,975 correctable adverse calls. 8 have zero modeled RE gain, leaving 4,967 positive-value actions for the exact-location oracle. Clubs captured 228.363 modeled RE with their 2,275 observed challenges. The comparison in Table 1 reports quantities per MLB game across both teams.

Table 1: Held-out performance on the factual opportunity clock

Policy	RE/game	Gain vs. observed [95% CI]	Model-implied overturn rate	Oracle share
Observed decisions	0.459	—	54.2%	32.8%
Learned policy	0.560	0.100 [0.074, 0.128]	45.3%	40.0%
Bellman check	0.561	0.101 [0.075, 0.129]	44.2%	40.1%
Exact-location oracle	1.399	0.940 [0.892, 0.987]	100.0%	100.0%

Notes: Values are modeled run expectancy, not realized runs. Brackets are paired intervals from 5,000 whole-game resamples of the held-out evaluation set, holding fitted policies fixed.

The learned policy captures 0.560 RE per game, improving on observed decisions by 0.100 (whole-game fixed-policy 95% interval 0.074–0.128) or 21.8%. Its model-implied overturn rate falls from 54.2% to 45.3%, but expected attempts rise from 2,275 to 2,961.7. Expected successful challenges still rise from 1,233 to 1,341. The policy is therefore less selective by accuracy and more selective by consequence.

Offense contributes 0.068 RE per game, or 68.1% of the total improvement; defense contributes 0.032. Offense produces the larger gain despite its wider effective action curve. Although offense has fewer adverse-call opportunities, positive-value correctable calls are more frequent (12.0% versus 4.4%), producing more oracle value (0.744 versus 0.655 RE/game) even though their mean value is lower.

The Bellman check reproduces the direct result (0.561 versus 0.560 RE/game), while public-information and fitted-imitation comparators reach only 0.196 and 0.427. Figure 1 shows why the learned rule grows more aggressive late: declining C_1 and C_2 lower the price of inventory as opportunities disappear.

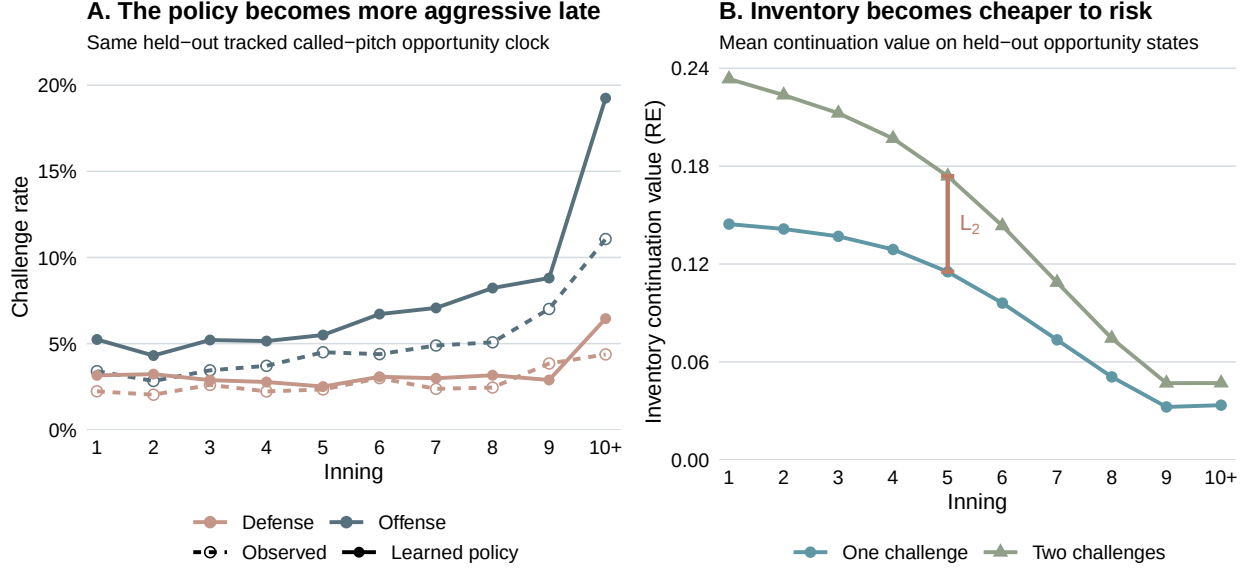


Figure 1: Challenge timing and inventory value in the held-out 497-game confirmation period. Panel A compares observed challenge rates with the learned policy’s expected rates by inning and role; innings after the ninth are pooled. Panel B shows the learned total continuation value of holding one (C_1) or two (C_2) challenges. The vertical bracket at inning five illustrates the marginal loss $L_2 = C_2 - C_1$. Falling continuation values make waiting less attractive late in games.

4.1 Two decisions at opposite stakes

Table 2 presents two held-out examples that make the threshold concrete.

Table 2: Two held-out challenge opportunities

Case	State	Role	G	L_2	q_2^*	$P(A M)$
Challenge (KC at LAD)	T1, 2 outs, 3-2	Offense	0.782	0.090	0.103	0.950
Wait (CLE at CIN)	B1, 2 outs, 0-0	Offense	0.032	0.085	0.727	0.008

Notes: $P(A = 1 | M = m, k = 2)$ is the *ex post* action probability obtained by integrating over the latent private signal conditional on realized geometry. The live decision statistic is $q = P(M > 0 | S, r, c)$.

The Kansas City example pairs a large full-count, runners-on gain with a modest failure cost, giving $q_2^* = 10.3\%$. In the Cleveland-at-Cincinnati example, a much smaller gain raises the threshold to 72.7%. A low-stakes pitch can therefore demand far higher confidence even when public geometry later says the call was wrong.

4.2 Descriptive team variation

Figure 2 reports the same held-out comparison by club. Arizona has the largest point estimate, whereas New York Yankees’s observed choices slightly outperform the learned policy in this window. Each club contributes only 31–35 games, as reflected in the wide whole-game intervals.

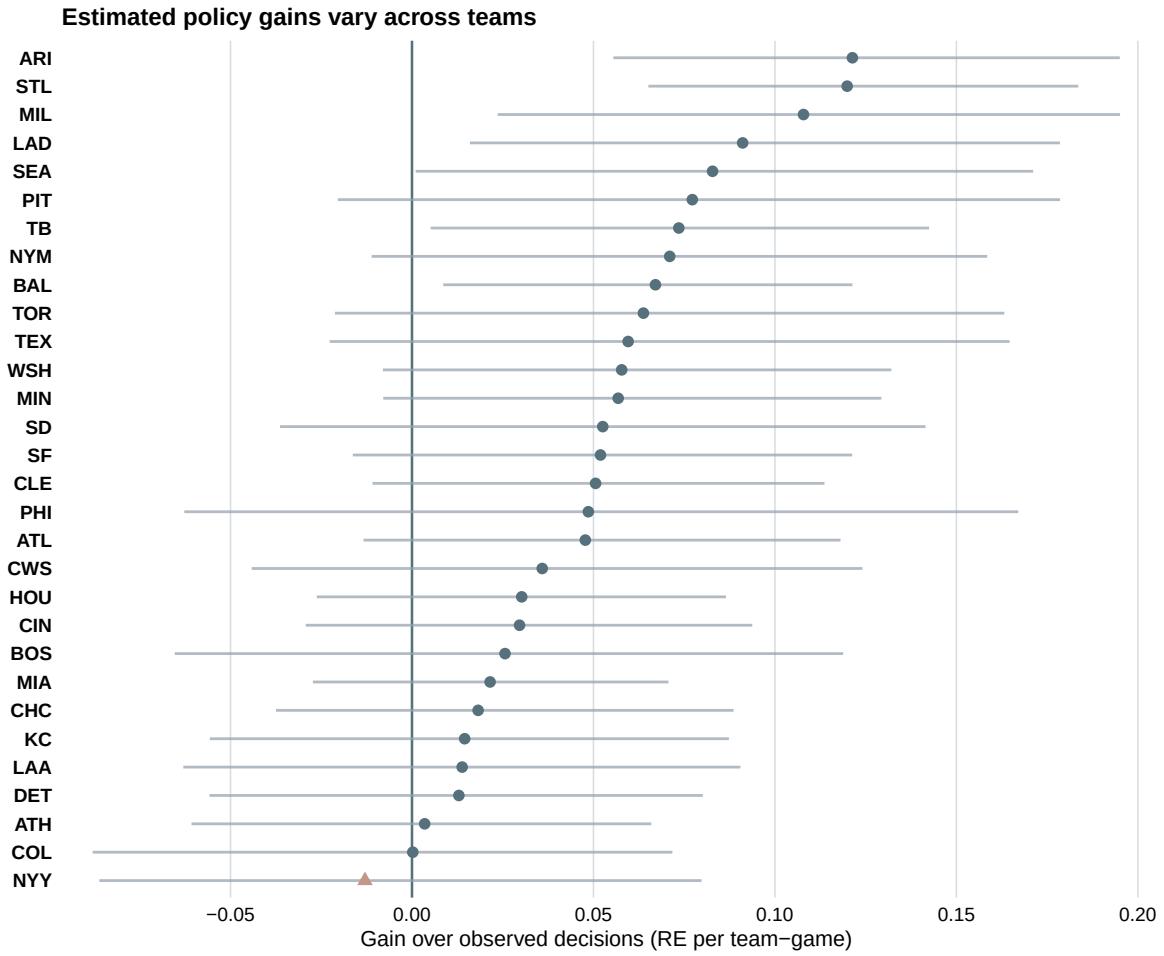


Figure 2: Estimated change in modeled RE per team-game under the learned policy relative to observed decisions. Points are team estimates; lines are 95% intervals from 5,000 whole-game resamples with the learned policy held fixed. The vertical line marks no change. Rankings are descriptive of this confirmation window.

5 Discussion

ABS policy should maximize expected run value, not overturn percentage. The learned rule lowers the required success probability when the current call is costly or future inventory has little remaining value, and raises it for low-value calls early in a game. The result is more modeled fixed-clock value despite more failed attempts. Offense has the wider action curve yet accounts for most of the estimated improvement, so effective discrimination alone does not determine policy value. The learned policy captures 40.0% of the hindsight, fixed-clock upper benchmark, leaving substantial room between a usable noisy-signal rule and hindsight. The practical implication is simple: challenge the calls that are worth being right about.

The estimand is modeled RE captured on the factual opportunity clock; downstream pitches remain at their observed states. Public tracking supplies the common adjudication standard, subject to measurement error. Observed actions identify role-level effective width; maximin selection handles its unidentified sensory-compliance split.

6 Reproducibility

The analysis code and reproducibility materials are available at <https://github.com/WhartonSABI/abs>. The repository contains locked R dependencies, tests, the compressed analysis inputs, the applied-results bundle, and the manuscript figure generator. MLB Stats API and Baseball Savant remain the authoritative upstream sources.

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